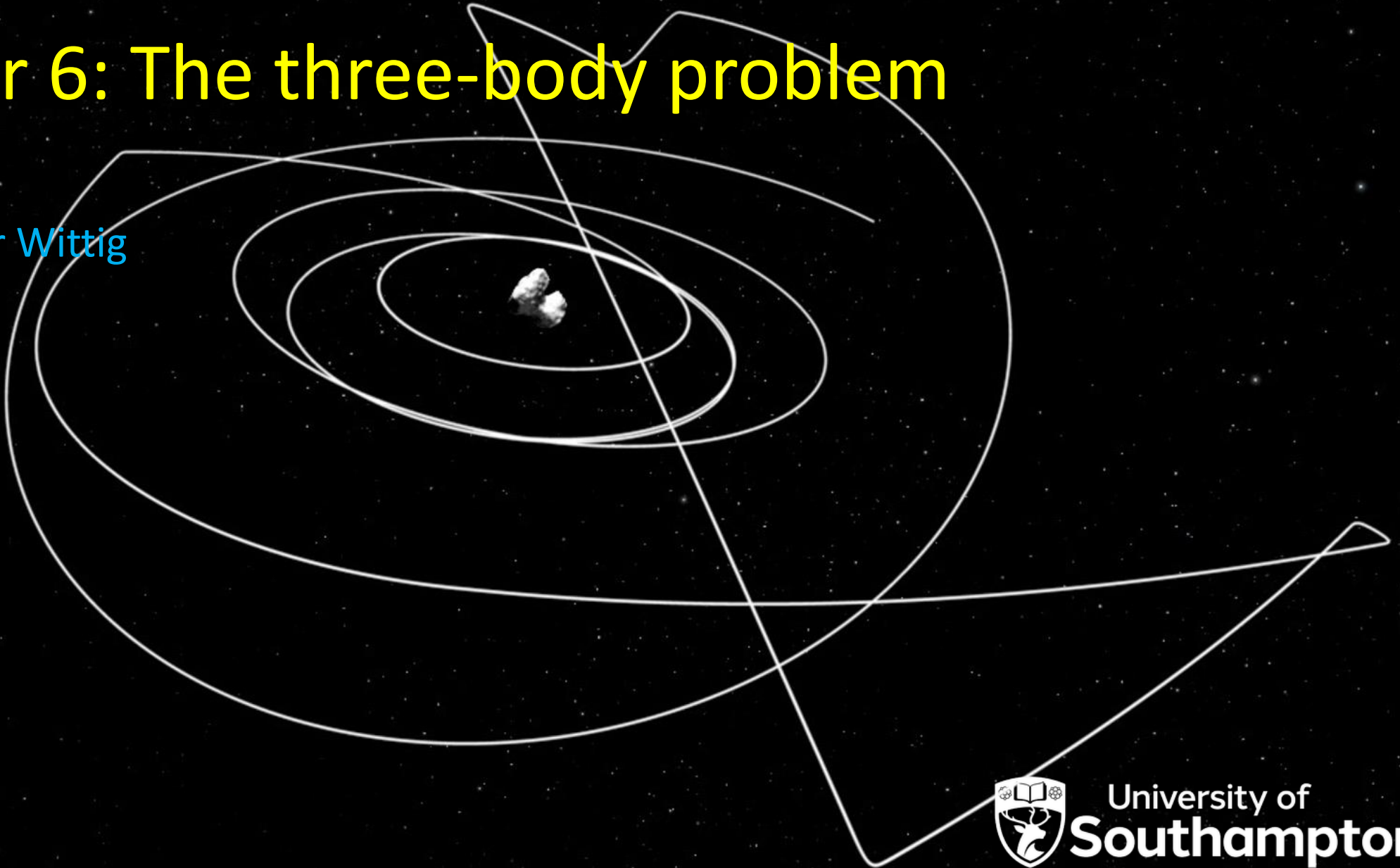


Orbital Mechanics (SESA6076)

Chapter 6: The three-body problem

Dr. Alexander Wittig



- Any questions on previous weeks content?

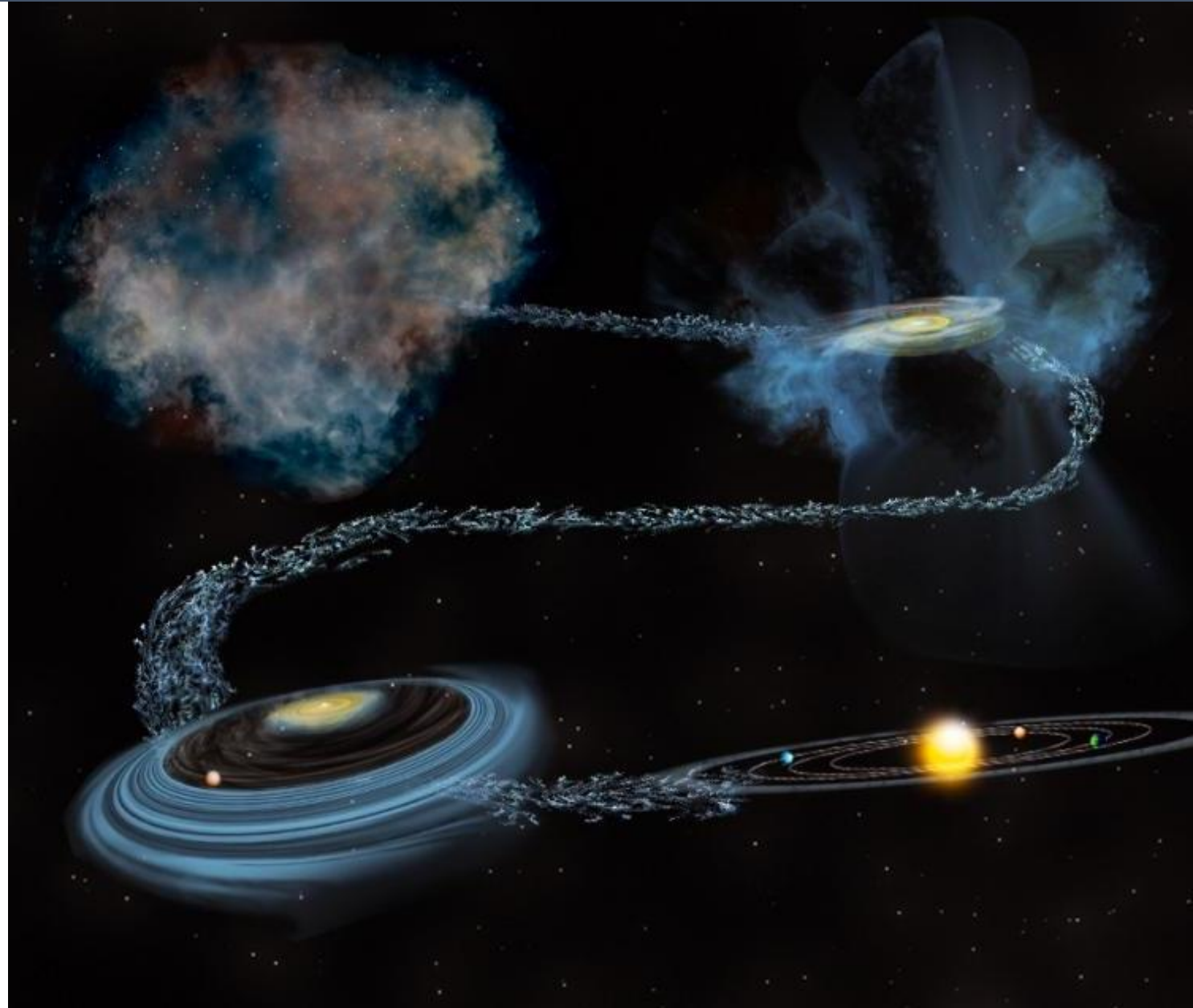
- **Introduction**
- **Chapter 1: In 20.000 km, turn right. Orbital Manoeuvres**
 - Coplanar Manoeuvres
 - Non-Coplanar Transfers
- **Chapter 2: That's mildly disturbing Orbital Perturbations & Propagation**
 - Non-Uniform Gravity Field
 - Third-Body Perturbation
 - Atmospheric Drag
 - Solar Radiation Pressure
 - Orbit Propagation Methods
- **Chapter 3: Is it a plane? Is it a bird? Orbit Observation & Determination**
 - Observation Techniques
 - Preliminary Orbit Determination
- **Chapter 4: Waltzing Matilda Relative Orbital Motion**
 - Relative Orbital Motion
 - Linearization & State Transition Matrix
- **Chapter 5: To boldly go... Interplanetary Trajectories & Optimization**
 - Lambert's Problem
 - Patched Conics
 - Gravity-assisted fly-by trajectories
- **Chapter 6: Torn between two bodies Restricted Three-Body Problem**
 - N-Body Problem
 - Circular Restricted Three-Body Problem
 - Natural Dynamics around Libration Points
- **Chapter 7: You spin me right round Attitude Dynamics**
 - Attitude representation
 - Attitude Dynamics
- **Chapter 8: In the spin cycle Attitude Motion**
 - Torque-Free Motion
 - Attitude Maneuvers
- **Summary and Revision**

- State the n-body problem, and derive the three-body, restricted three-body, and circular restricted three-body problem
- Explain what the synodic reference frame is and where it comes from
- Identify and name Lagrange points
- Apply the concept of “forbidden regions” and derive them from a conserved energy
- Use 3-body dynamics for mission design purposes

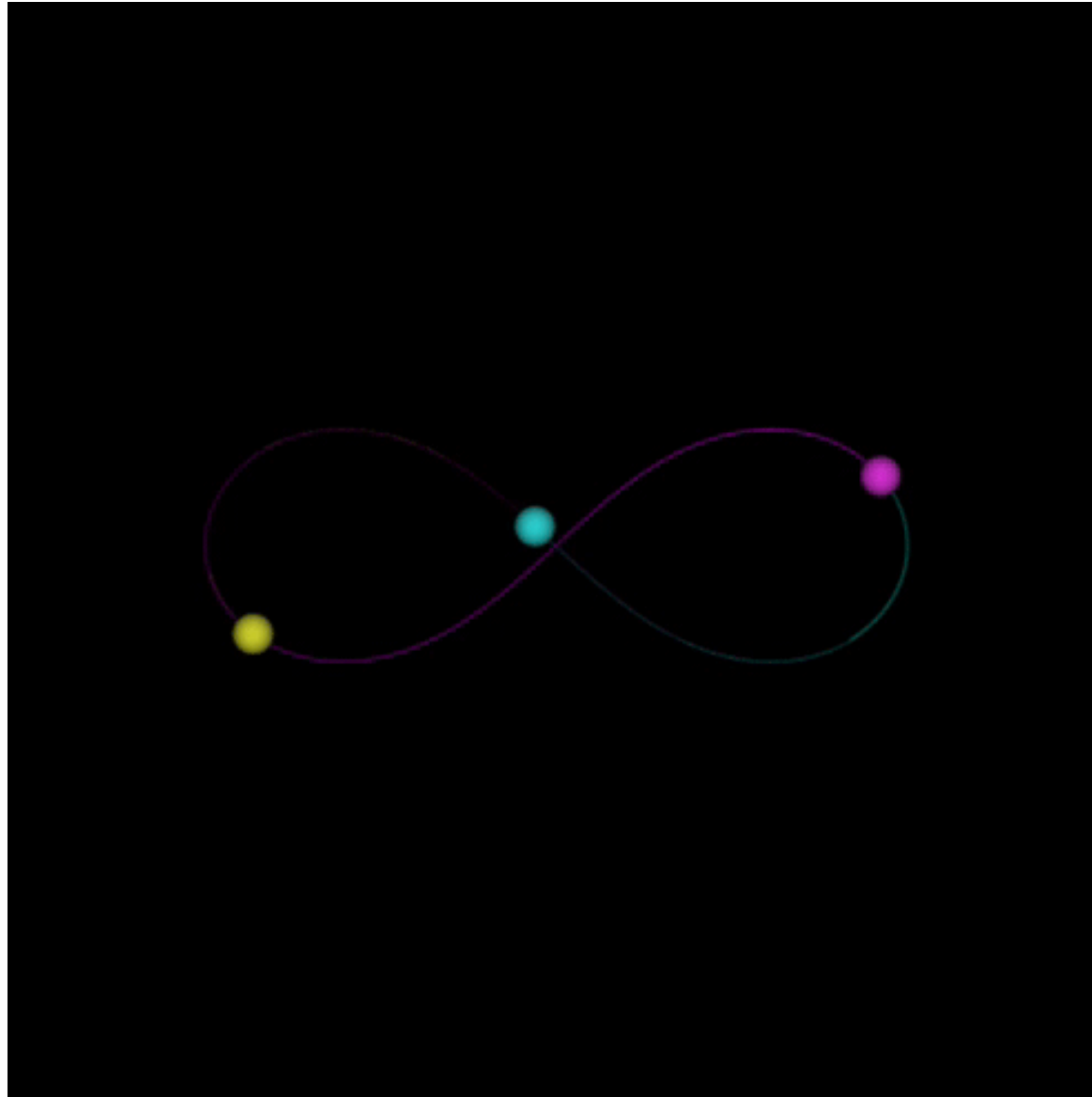
- Motion of n bodies under mutual gravitation
- Complex dynamics
- Computationally intensive

Applications:

- Solar System Formation
- Galactic dynamics
- High fidelity spacecraft dynamics



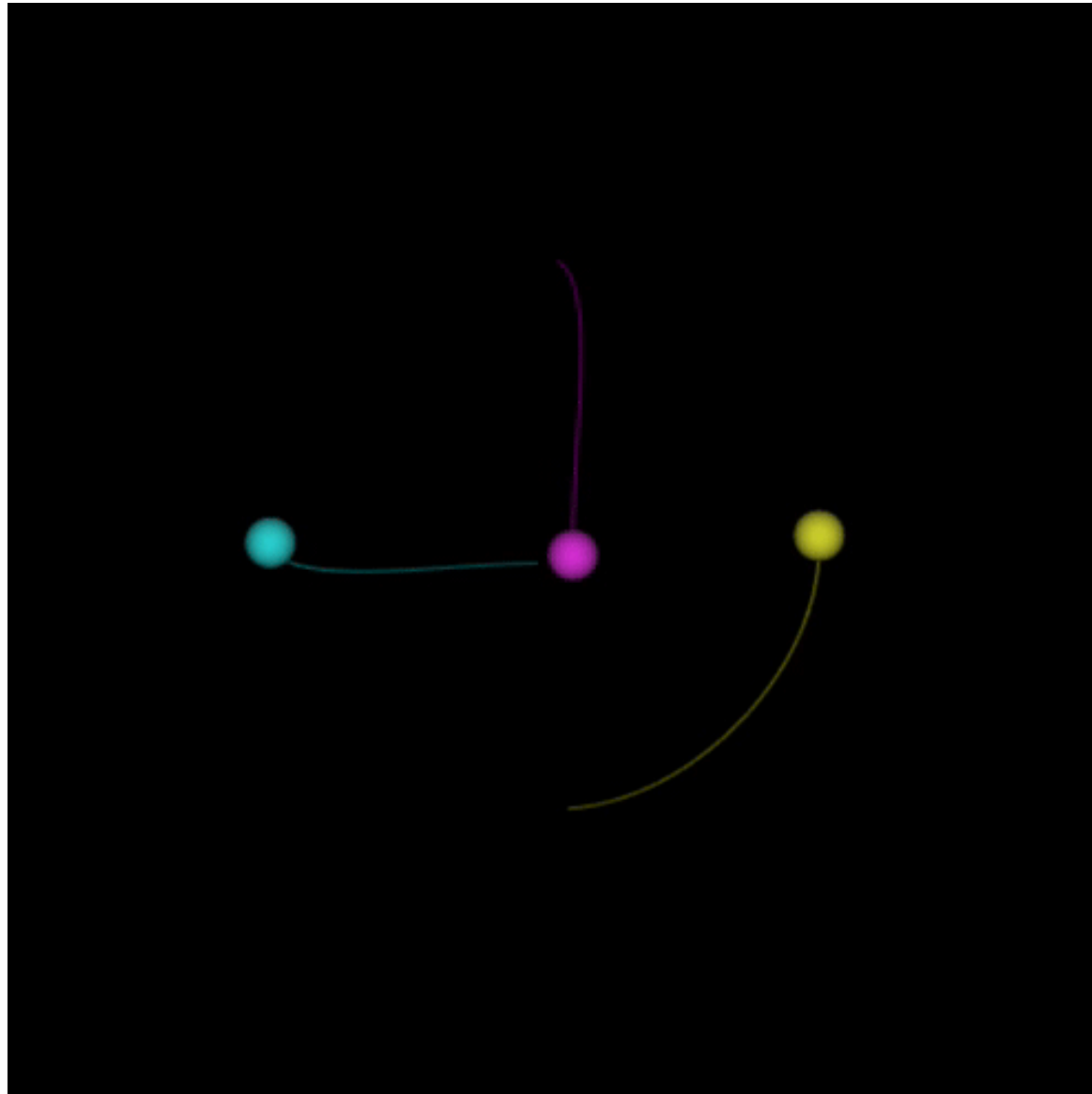
Some fun n-body trajectories



<https://blogs.scientificamerican.com/guest-blog/5-gifs-of-n-body-orbits-animation/>

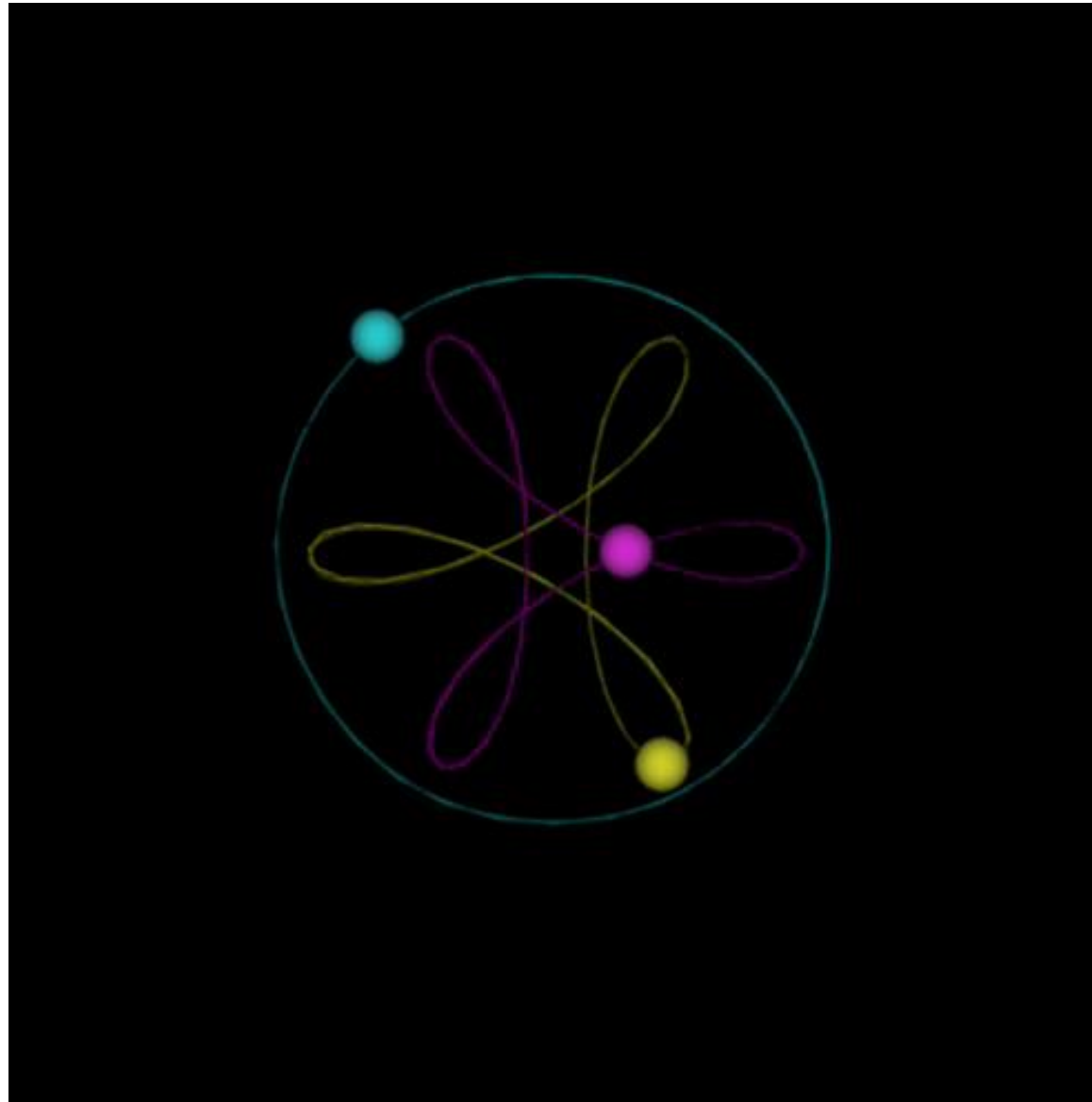
Chris Moore

Some fun n-body trajectories



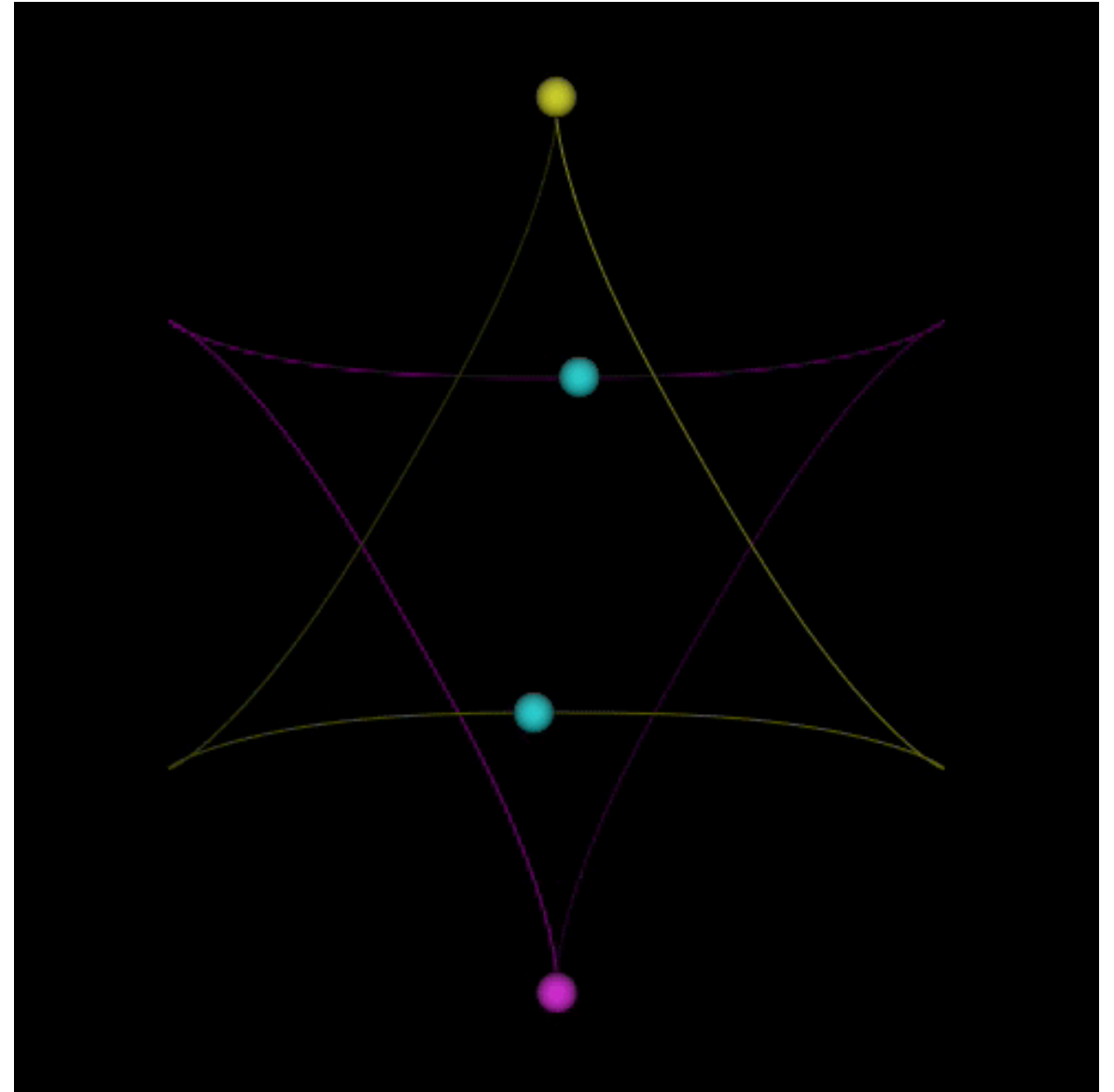
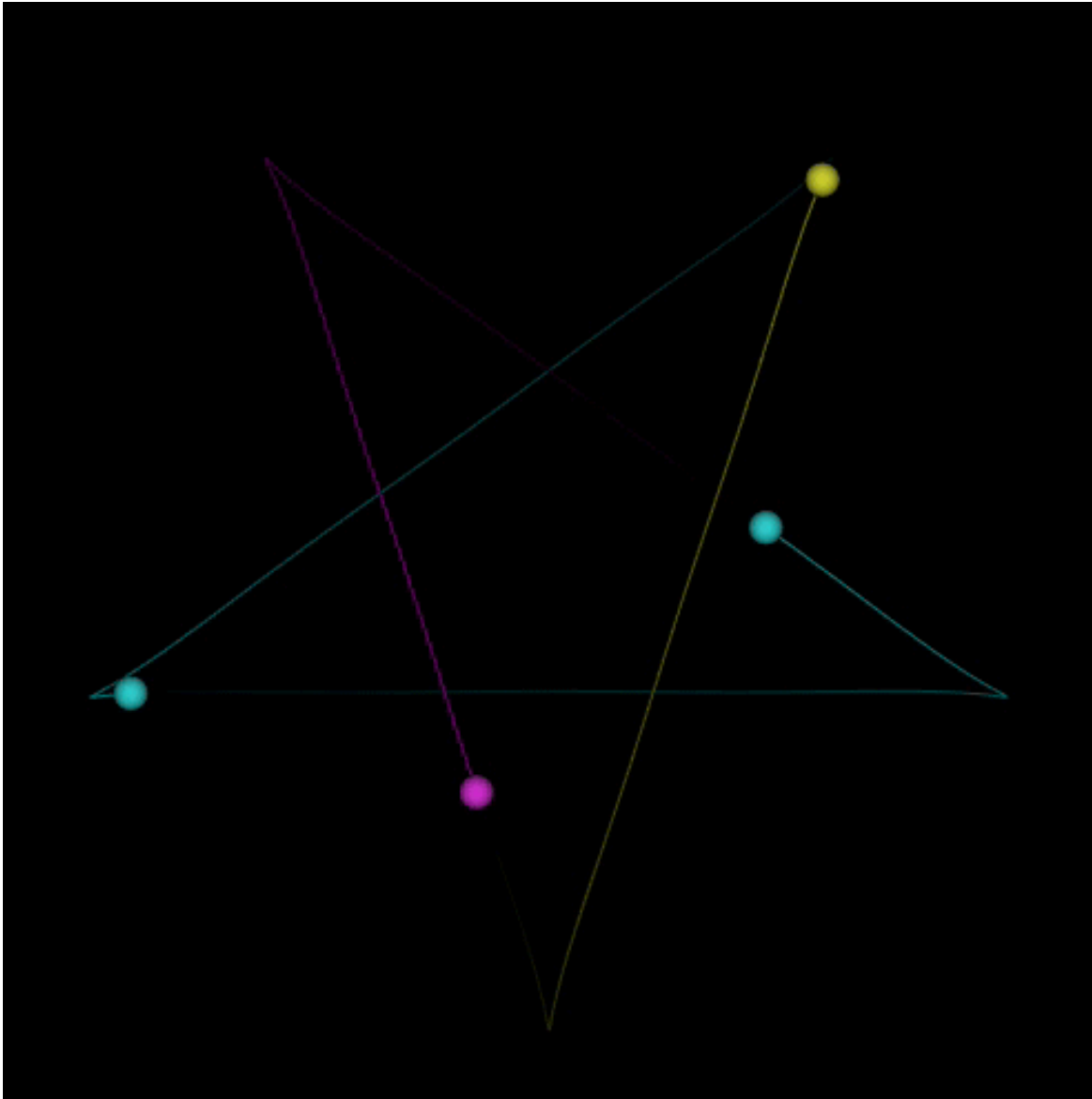
<https://blogs.scientificamerican.com/guest-blog/5-gifs-of-n-body-orbits-animation/>

Robert Vanderbei



<https://blogs.scientificamerican.com/guest-blog/5-gifs-of-n-body-orbits-animation/>

Some fun n-body trajectories



<https://blogs.scientificamerican.com/guest-blog/5-gifs-of-n-body-orbits-animation/>

Xie and Ouyang

Massive n-body: Solar System Formation

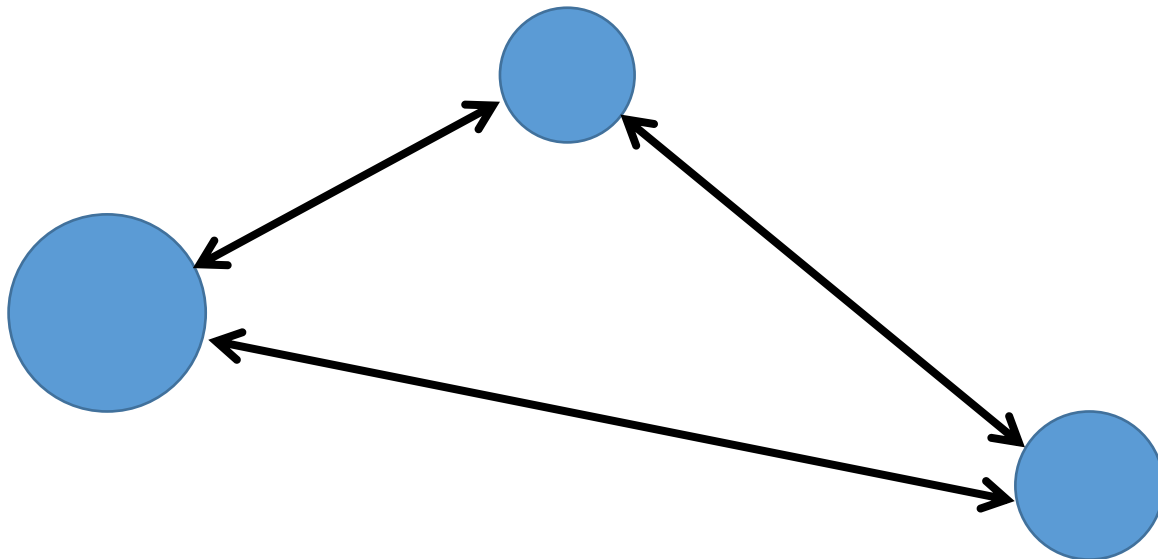
California Academy of Science

<https://www.youtube.com/watch?v=yXq1i3HlumA>

Three Body Problems

Three body problem:

- motion of **three bodies** under their mutual gravity
- Center of mass moves at constant velocity
 - valid for any n-body, also 2-body (Kepler)



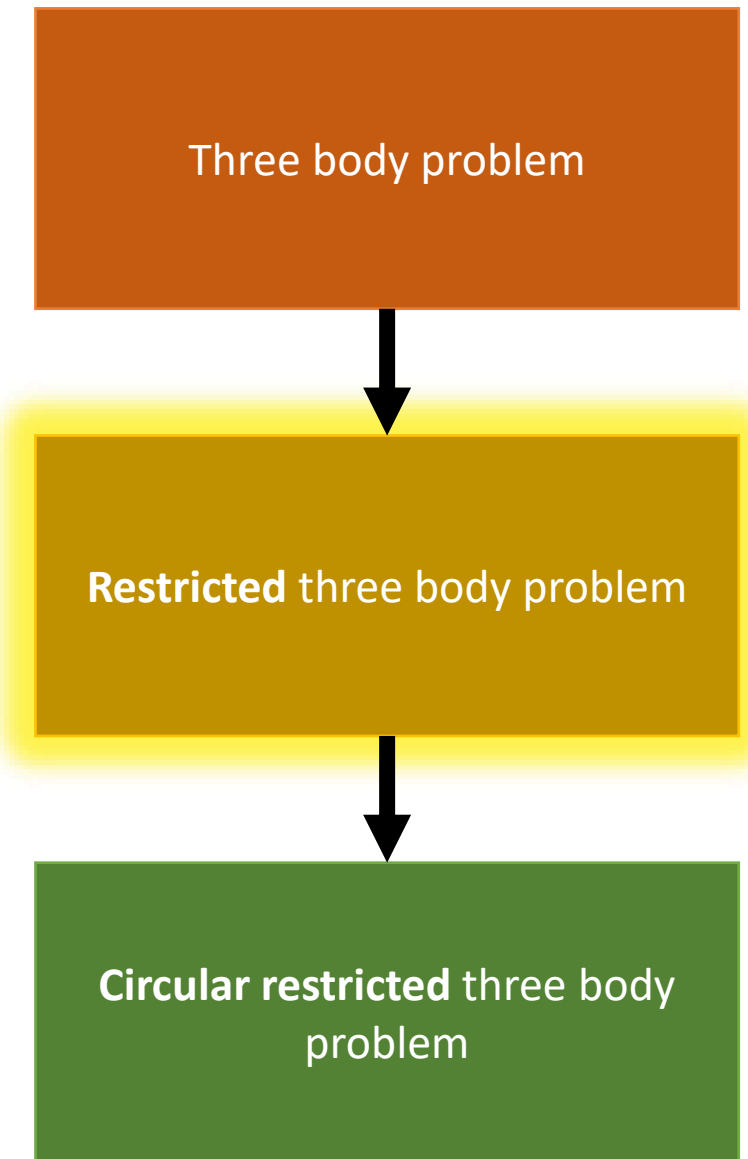
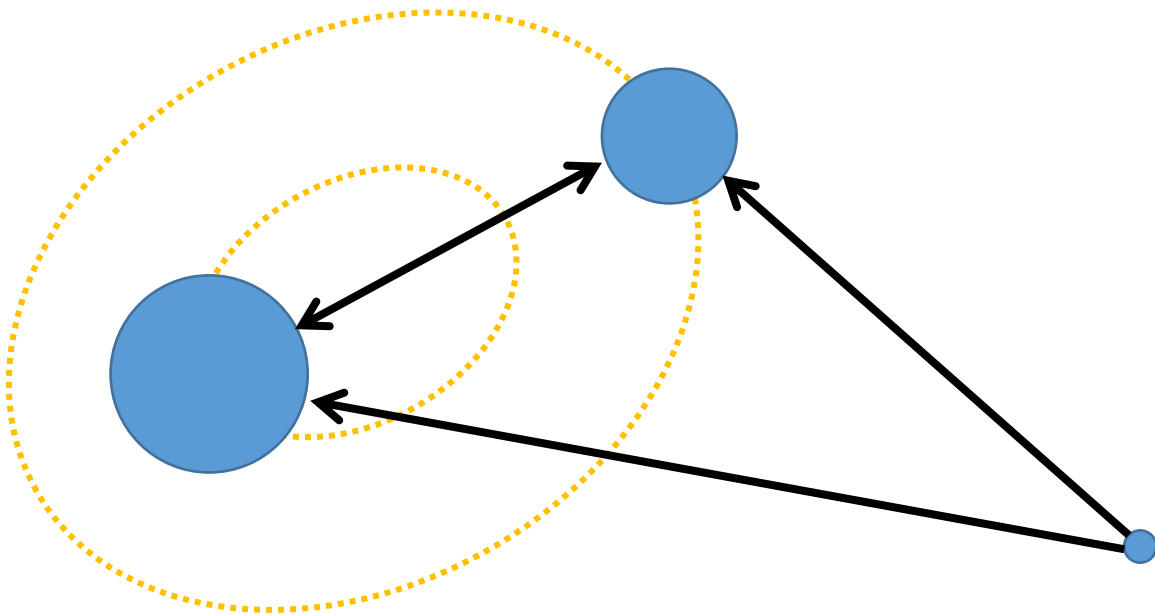
Three body problem

Restricted three body problem

Circular restricted three body problem

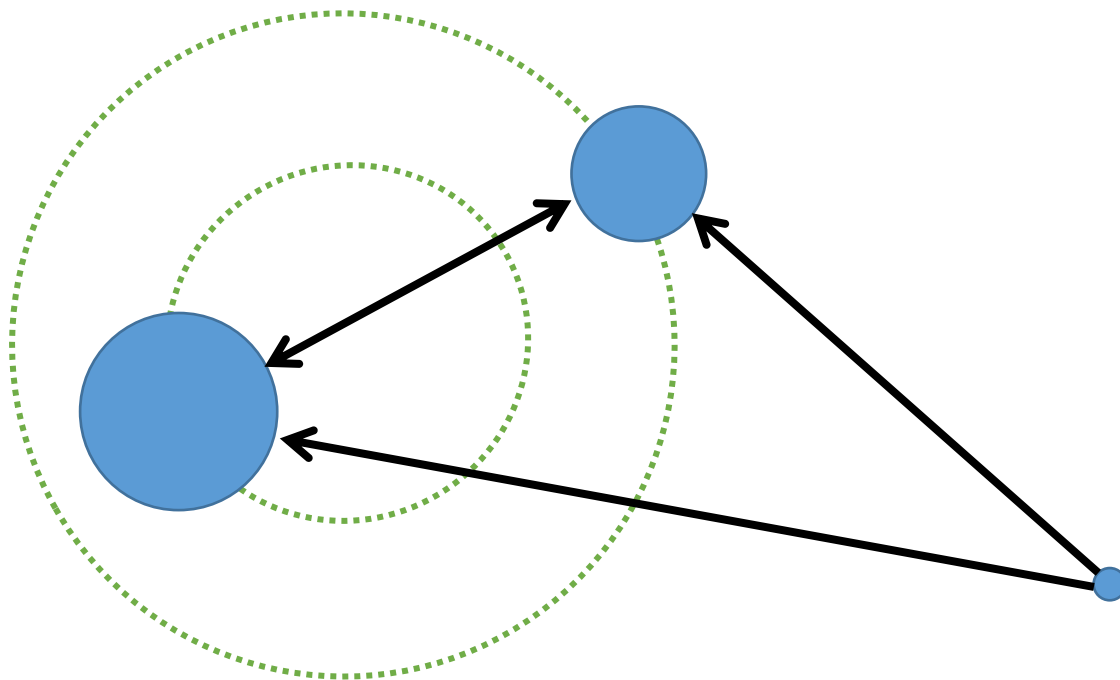
Restricted three body problem:

- motion of **two bodies** under Keplerian motion
- tiny third body moves in their field **without affecting them**



Circular restricted three body problem:

- motion of **two bodies** in **circular Keplerian orbits** around center of mass
- tiny third body moves in their field **without affecting them**



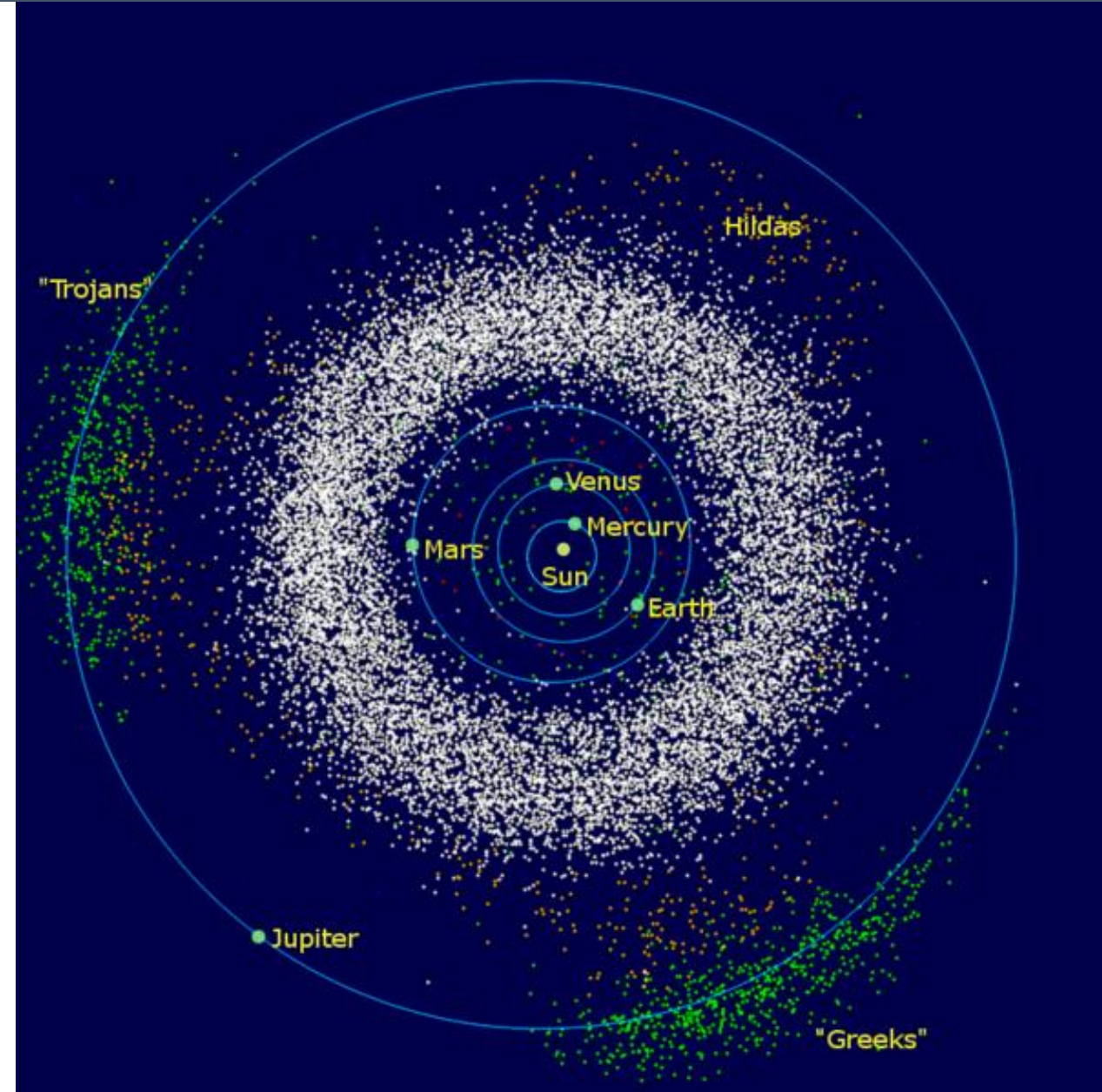
Three body problem

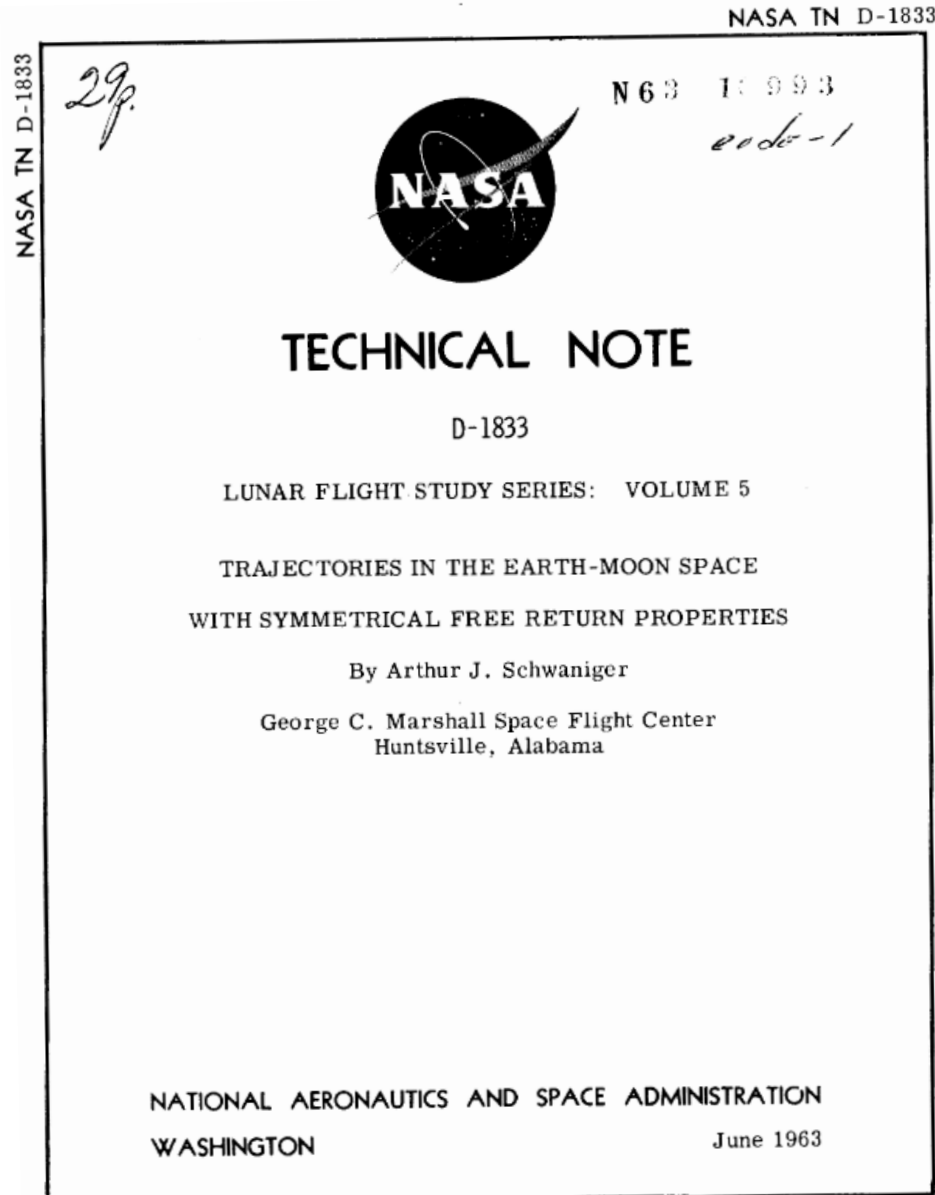
Restricted three body problem

Circular restricted three body problem

Examples of 3BP in solar system

- Earth-Moon
 - Patched conics not good enough for moon mission design & analysis
- Sun-Earth
 - Near Earth Asteroids
 - Solar System observation missions
- Sun-Jupiter
 - Essentially our solar system
 - Explains asteroid clusters





- Apollo lunar “figure 8” free return trajectories
- Three body dynamics allow closed orbit
- Without lunar orbit insertion returns automatically to reentry condition
- Safety for human space flight missions

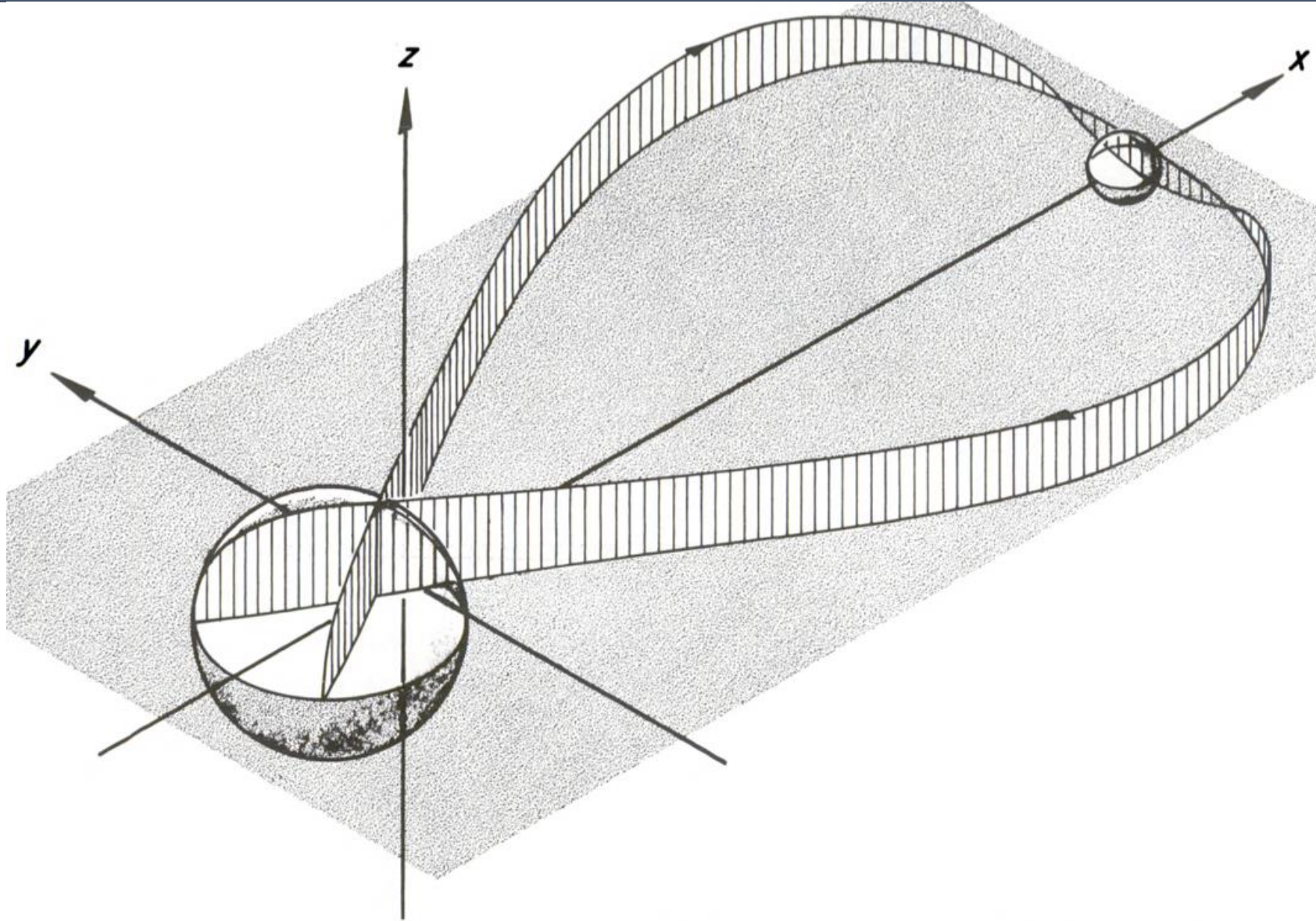


FIG. 4. FREE RETURN TRAJECTORY OF THE SECOND KIND

Drawn to Scale

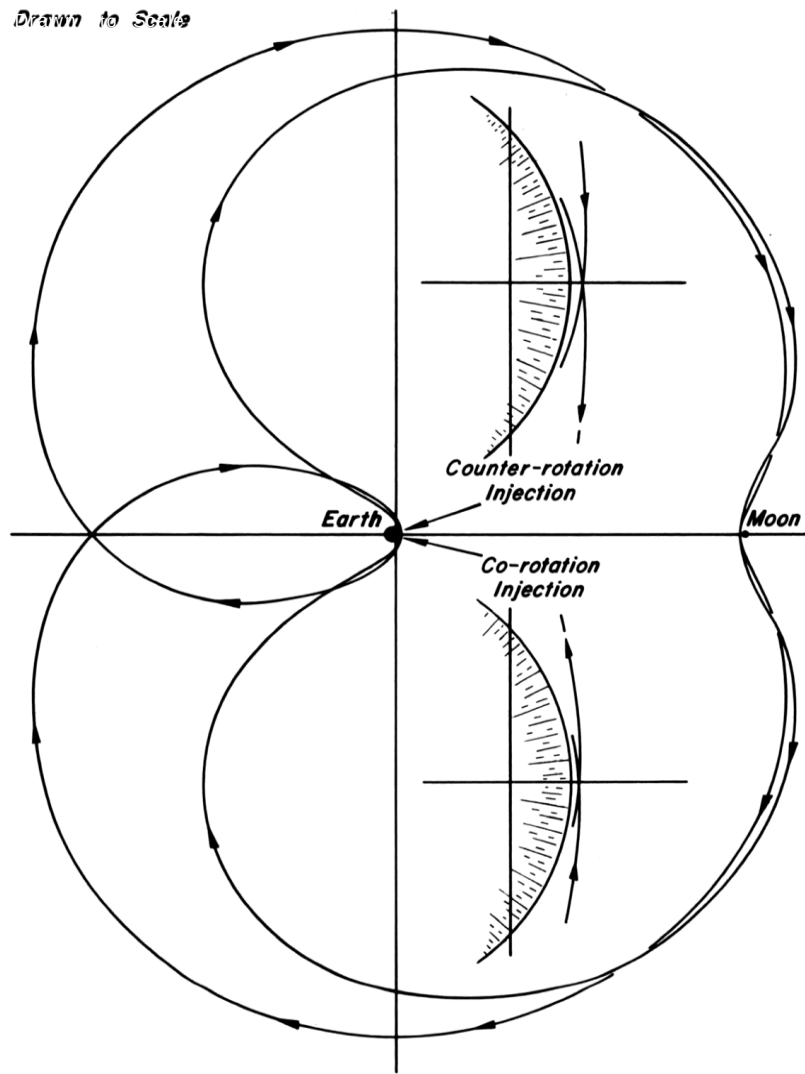


FIG. 6.
COPLANAR CISLUNAR FREE RETURN TRAJECTORIES
(Reference Frame Rotating with Moon.)

Drawn to Scale

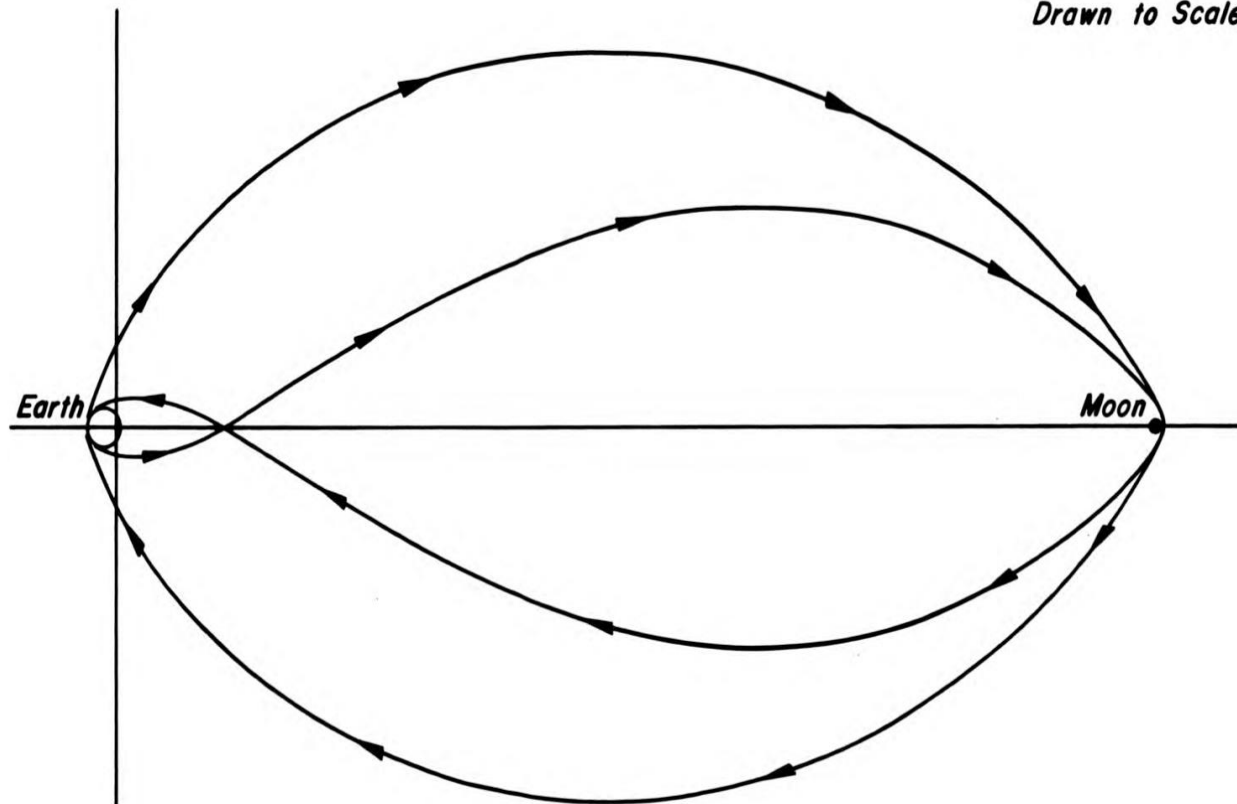


FIG. 5. COPLANAR CIRCUMLUNAR FREE RETURN TRAJECTORIES
(Reference Frame Rotating with Moon)

Career:

- Research Mathematician, “Computer” (NACA, 1953)
- Aerospace technologist (NASA, 1958)

Achievements:

- Manual (!) trajectory and launch window calculations for Mercury mission (1961)
- Verification of first digital computers for Mercury 6 (1962)
- Designed back up procedures for Apollo 13 allowing manual navigation (1970)



Circular Restricted Three Body Problem (CR3BP)

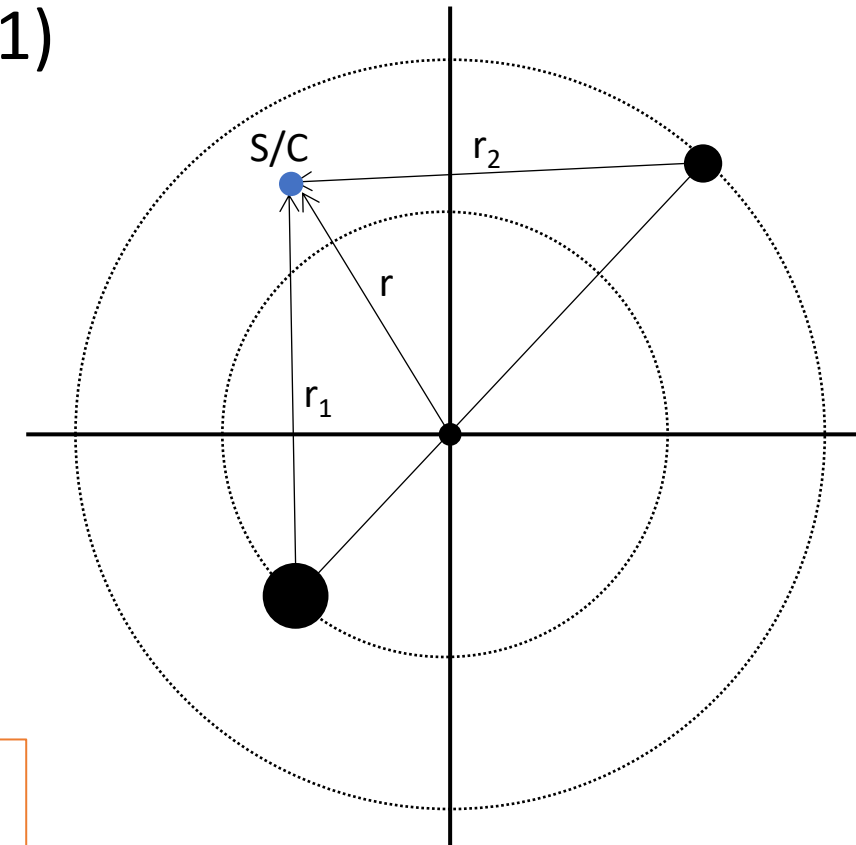
- Origin: center of mass
- 2 primaries on circular orbits around CM (Chapter 1)
- Normalized units (for simplicity)
- System parameter: mass ratio

$$\mu = \frac{m_2}{m_1 + m_2}$$

Not to be confused with gravitational
parameter in Keplerian motion!

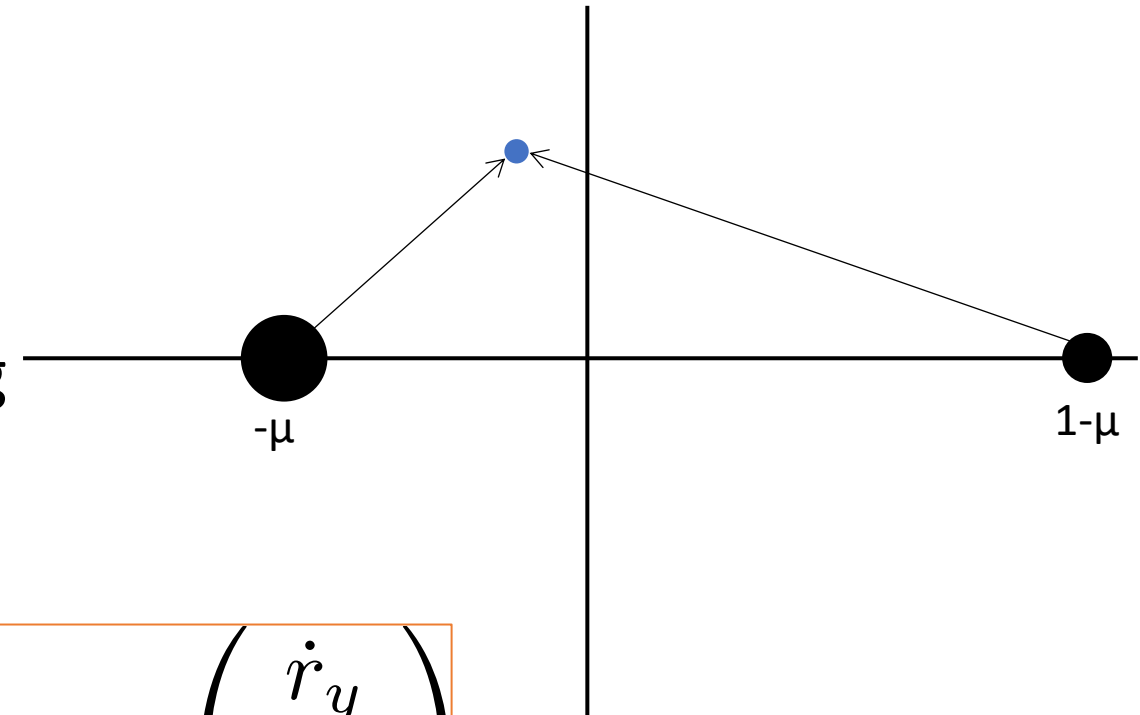
Example Earth-Moon: $\mu = 0.0121505$

Problem: non-autonomous
(= right hand side depends explicitly on time)



Rotate frame around center of mass with angular velocity Ω

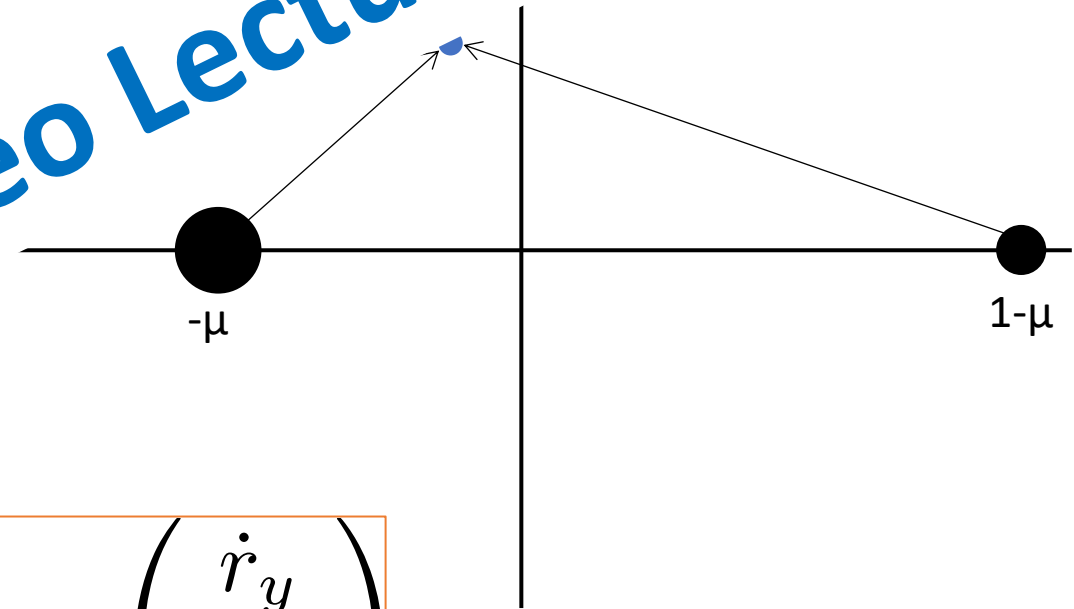
- Primaries become stationary
- Frame not inertial any more
- Equations of motion change by switching into rotating frame
- No more explicit time dependence!



$$\ddot{\vec{r}} = -\frac{1-\mu}{|\vec{r}_1|^3}\vec{r}_1 - \frac{\mu}{|\vec{r}_2|^3}\vec{r}_2 + \begin{pmatrix} r_x \\ r_y \\ 0 \end{pmatrix} + 2 \begin{pmatrix} \dot{r}_y \\ -\dot{r}_x \\ 0 \end{pmatrix}$$

Rotate frame around center of mass with angular velocity Ω

- Primaries become stationary
- Frame not inertial any more
- Equations of motion change into rotating frame
- No more explicit reference!



$$\ddot{\vec{r}} = -\frac{1}{|\vec{r}_1|^3} \vec{r}_1 - \frac{\mu}{|\vec{r}_2|^3} \vec{r}_2 + \begin{pmatrix} r_x \\ r_y \\ 0 \end{pmatrix} + 2 \begin{pmatrix} \dot{r}_y \\ -\dot{r}_x \\ 0 \end{pmatrix}$$



Jacobi constant

$$\begin{aligned}C_J &= -2E \\&= 2\frac{Gm_1}{|\vec{r}_1|} + 2\frac{Gm_2}{|\vec{r}_2|} + \Omega^2(r_x^2 + r_y^2) - (\dot{r}_x^2 + \dot{r}_y^2 + \dot{r}_z^2) \\&= 2\left(\frac{1-\mu}{|\vec{r}_1|} + \frac{\mu}{|\vec{r}_2|}\right) + r_x^2 + r_y^2 - \dot{r}_x^2 - \dot{r}_y^2 - \dot{r}_z^2\end{aligned}$$

- Only (known) constant of motion in CR3BP



Jacobi constant

$$C_J = -2E$$

$$= 2Gm$$

$$\left(\frac{-\mu}{|\vec{r}_1|} + \frac{\mu}{|\vec{r}_2|} \right) + r_x^2 + r_y^2 - \dot{r}_x^2 - \dot{r}_y^2 - \dot{r}_z^2 + \Omega^2(r_x^2 + r_y^2) - (\dot{r}_x^2 + \dot{r}_y^2 + \dot{r}_z^2)$$

- Only (known) constant of motion in CR3BP

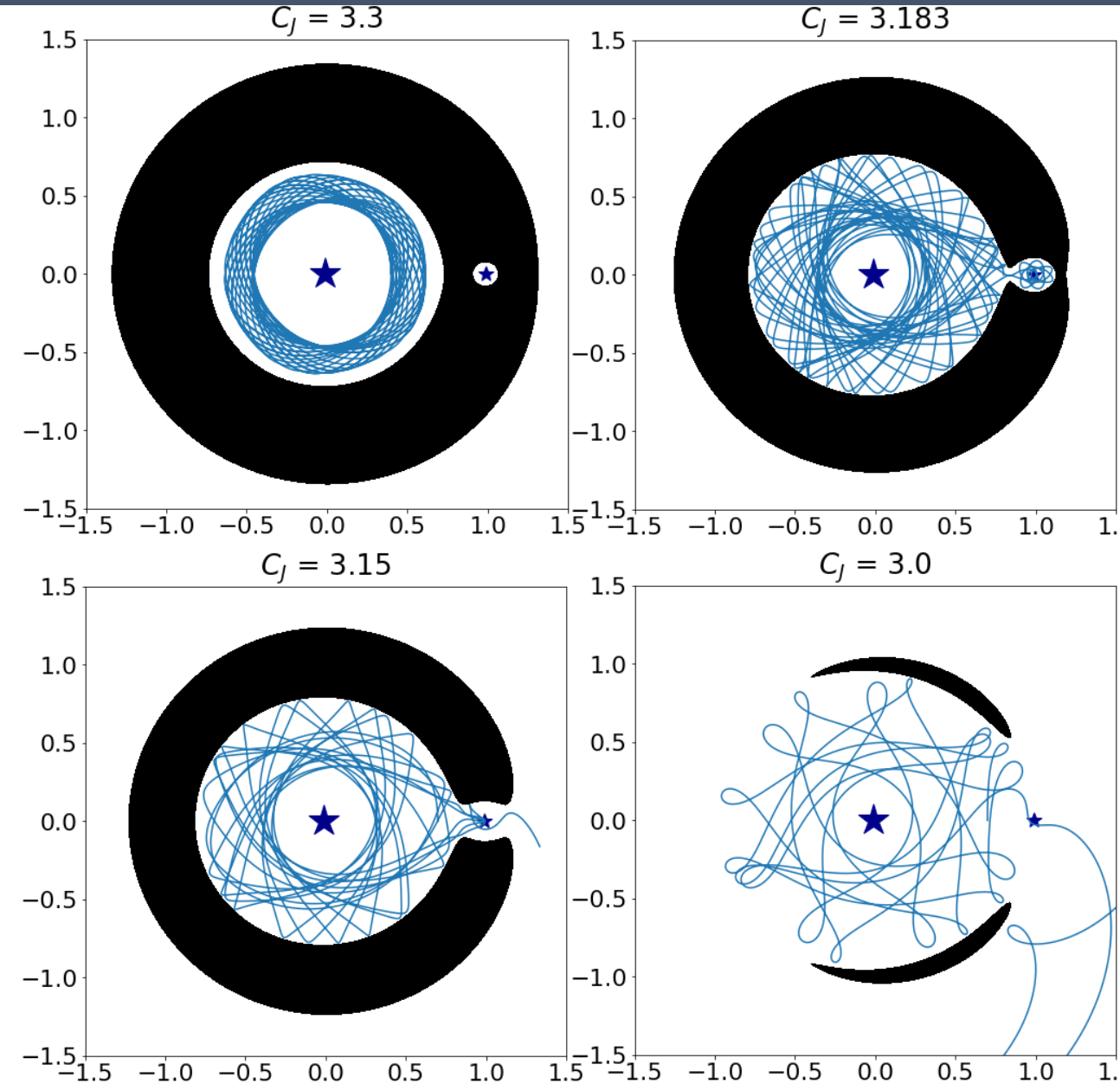
Applications of zero velocity curves

Forbidden regions

- Depend on
 - μ (system)
 - C_J (spacecraft)
- Physically impossible to occupy
- Same idea as energy surfaces

Applications

- Planetary protection
- Spacecraft disposal:
 - Increase energy to open region
 - Move S/C outside
 - Close zero-velocity curves to lock out disposed spacecraft



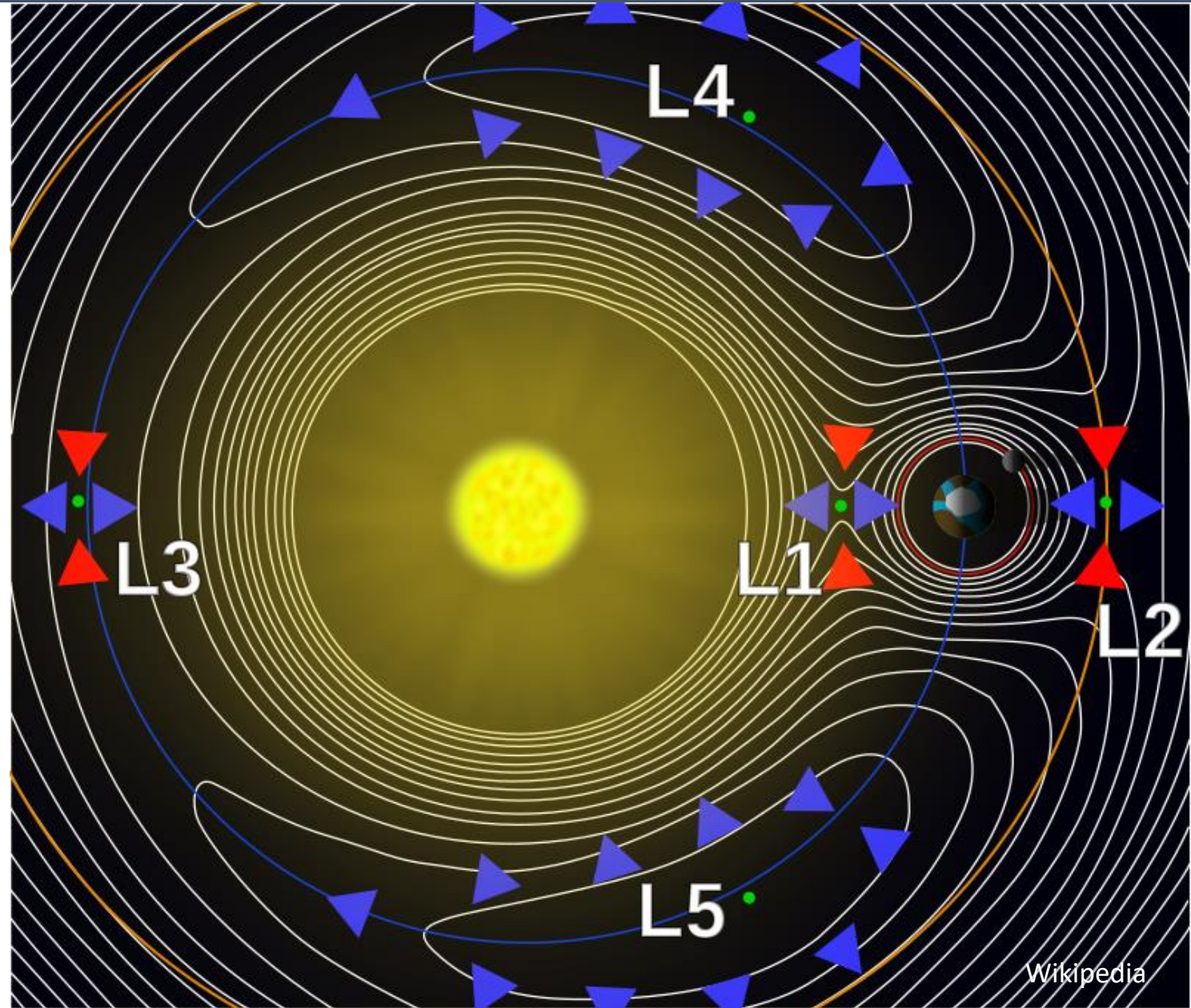
Lagrange Points

Lagrange points

- Fixed points as example of dynamical features
- All forces cancel
- S/C with zero velocity stays there
- L1, L2, L3 – unstable
- L4, L5 – stable
if $m_1/m_2 > 24.96$
 - true for most CR3BP in our solar system

Other dynamical features:

- Periodic orbits
- Invariant manifolds



Wikipedia

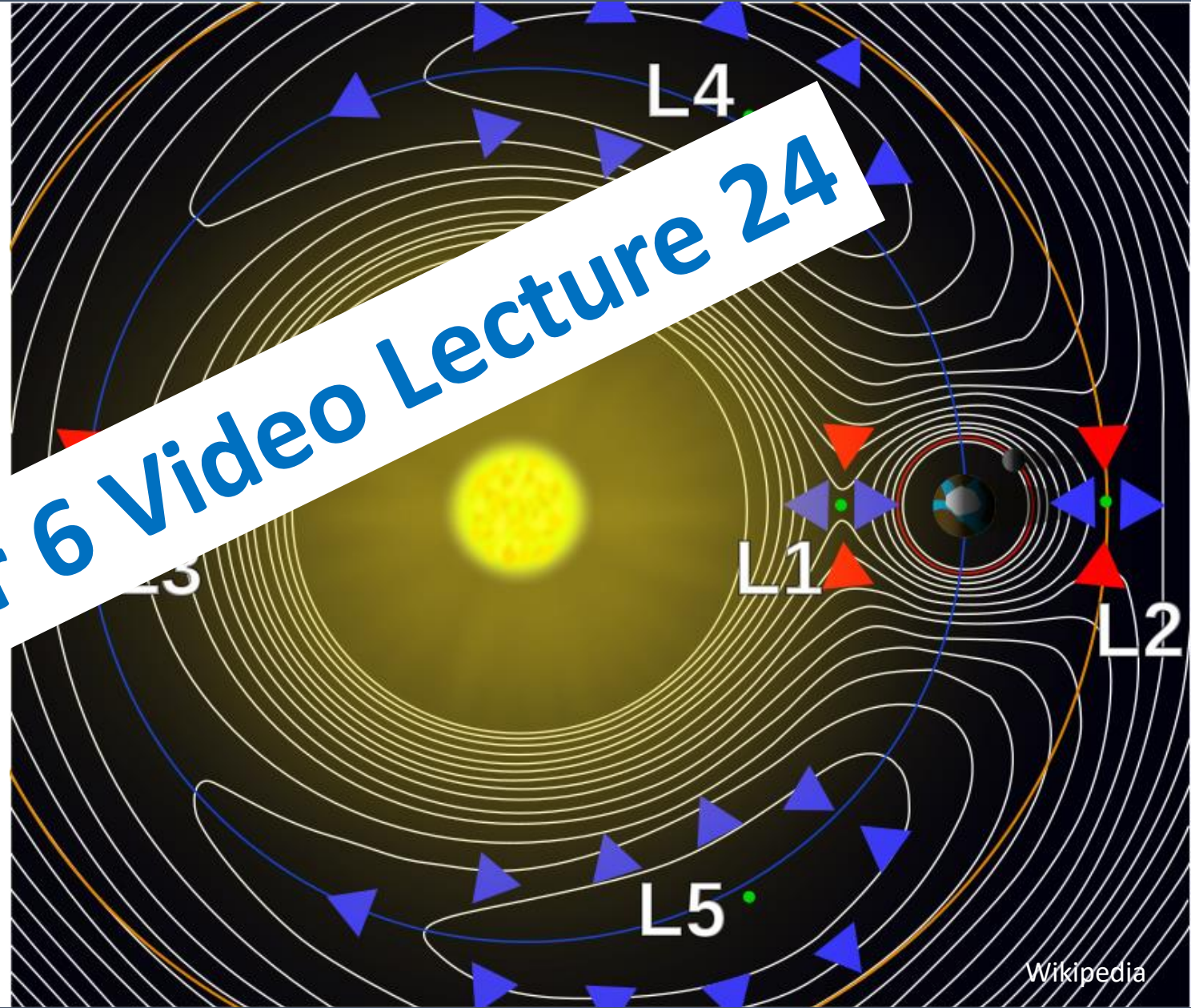
Lagrange points

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Other dynamical features:

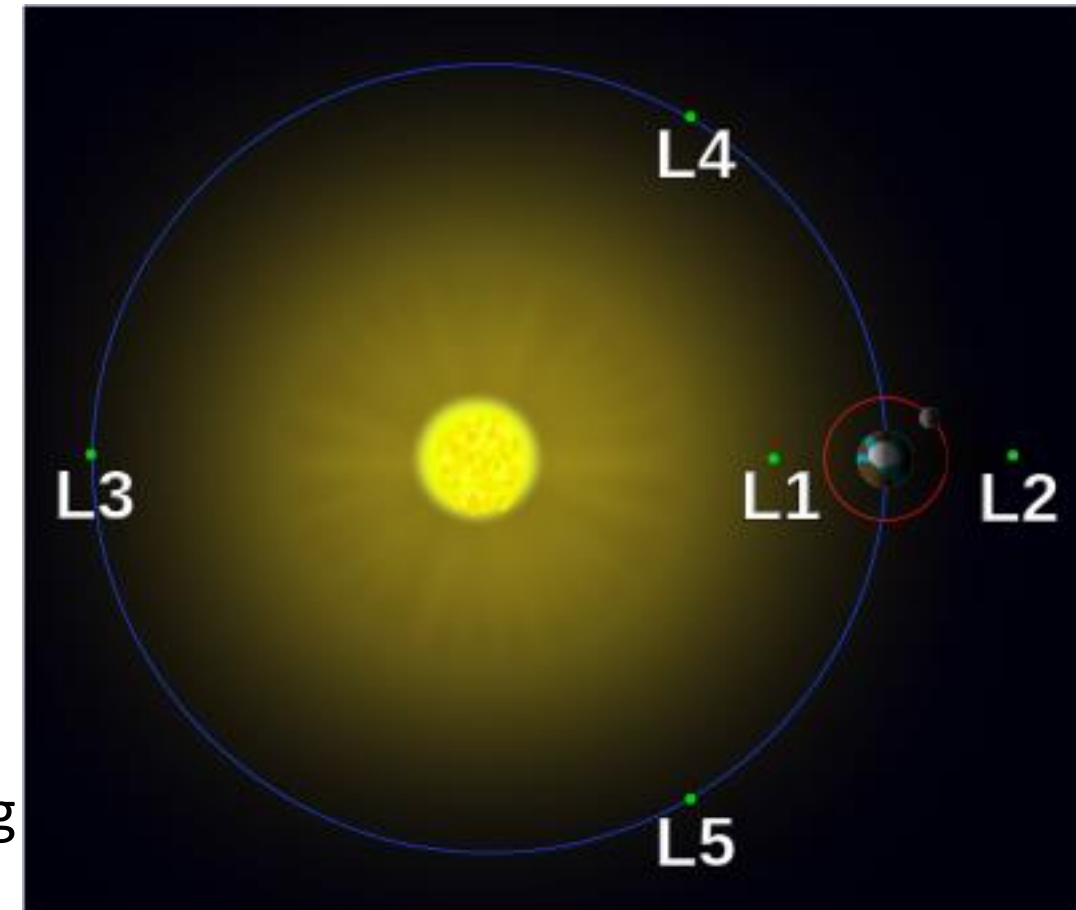
- Periodic orbits
- Invariant manifolds

see Chapter 6 Video Lecture 24



Wikipedia

- SEL1: Sun/Earth observation
 - always sees sun-lit side of Earth
 - never shadowed by Earth or Moon
 - Examples: Sun and Space weather observatories (SOHO, ACE)
- SEL2: Space telescopes
 - fixed position simplifies calibration & shielding
 - partly shaded by Earth
 - Examples: James Webb Telescope, GAIA
- SEL3: Sun observation
 - observing other side of Sun allows early warning
 - No missions yet
- SEL4/5: No missions yet



- N-body problem (in general) chaotic
- Simplifications: 3 bodies, restricted, circular restricted
- Applications in solar system
- CR3BP mathematical formulation
 - Synodic frame
 - Jacobi constant
- Zero velocity curves
- Lagrange points and applications

Further reading

- Vallado, Chapter 12.7

Orbital Mechanics (SESA6076)

Chapter 6: The three-body problem

Dr. Alexander Wittig

